
Predicting Patient Care Plans from FHIR Data: A Comparison of LSTM and Transformer Models

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Abstract

We compare a time-aware Transformer and a multi-layer LSTM for predicting patient care plans from synthetic FHIR data generated by Synthea ($\sim 1,000$ patients). The Transformer encodes medical events into a unified token vocabulary with continuous time embeddings and uses self-attention to capture long-range clinical dependencies, while the LSTM combines one-hot encounter features with patient demographics via sequential processing. The Transformer substantially outperforms the LSTM, achieving 70% Top-1 accuracy with macro precision/recall/F1 of 0.75/0.70/0.66 versus the LSTM’s 0.10/0.20/0.11. This gap stems from self-attention’s ability to directly attend to any event in the patient timeline, whereas the LSTM’s hidden state progressively overwrites older clinical signals. We further demonstrate clinically meaningful interpretability via Integrated Gradients attribution and discuss directions for ensemble strategies and real-world deployment.

1 Introduction

The digitization of healthcare records through standards such as FHIR [1] has created opportunities for machine learning–driven clinical decision support. Care plan prediction, recommending appropriate treatment plans from a patient’s medical history, can reduce physician cognitive load and accelerate treatment initiation. However, EHR data presents modeling challenges: variable-length encounter sequences, heterogeneous event types, irregular time intervals, and large medical code vocabularies.

Problem. Rule-based clinical decision support cannot adapt to the complexity of real patient trajectories. LSTMs [3] have been widely used for sequential clinical data but struggle with long-range dependencies spanning hundreds of encounters. Transformer architectures [4] avoid this bottleneck via self-attention, yet limited work directly compares LSTMs and Transformers for care plan prediction from structured FHIR data.

Approach. We implement and compare: (1) a **Time-Aware Clinical Transformer** that embeds medical event codes and continuous time deltas into a unified representation with self-attention over the full timeline, and (2) a **Multi-Layer LSTM** combining one-hot encounter features with static patient demographics. Our Transformer significantly outperforms the LSTM on all metrics while providing event-level interpretability through Integrated Gradients [10].

Contributions. (1) A reproducible FHIR preprocessing pipeline, (2) a rigorous architecture comparison for care plan prediction, (3) evidence that attention-based models better capture long clinical histories, and (4) an interpretability analysis using Captum [9].

2 Background and Related Work

EHRs and FHIR. The FHIR standard [1] provides a RESTful framework for representing healthcare data using resources such as Patient, Encounter, Condition, and CarePlan. Synthea [2] generates realistic FHIR-compliant synthetic patient records, enabling ML research without privacy concerns.

Deep Learning for Clinical Prediction. LSTMs [3] have been widely applied to sequential EHR data. Doctor AI [5] used GRUs to predict diagnoses and medications, while RETAIN [6] added attention over RNN outputs for interpretability. The Transformer [4] avoids sequential processing via self-attention. BEHRT [7] adapted BERT for structured EHR diagnosis prediction, and MedBERT [8] pre-trained Transformers on large-scale EHR data. These works demonstrate Transformers’ effectiveness for temporal clinical patterns.

Care Plan Prediction and Interpretability. Care plan prediction requires understanding the holistic patient context, the interplay of conditions, medications, observations, and demographics, to recommend a treatment strategy, making it more complex than diagnosis prediction. Clinical AI deployment further requires interpretability [11]; methods such as Integrated Gradients [10] and SHAP [12] quantify per-feature contributions to predictions. While both LSTM and Transformer architectures have been applied to various EHR tasks, a direct comparison for care plan prediction using structured FHIR data with unified event vocabularies and time-aware embeddings is lacking. This work addresses that gap.

3 Methods

3.1 Dataset

We use the Synthea [2] synthetic patient dataset (April 2020 CSV release) containing $\sim 1,000$ patients with complete medical histories spanning allergies, care plans, conditions, encounters, immunizations, medications, observations, and procedures, each coded using standard terminologies (SNOMED-CT, RxNorm, CVX, LOINC).

3.2 Data Preprocessing

Transformer Pipeline. We construct a unified event timeline per patient. Six clinical CSV tables are loaded and standardized; domain-specific prefixes (COND_, MED_, OBS_, etc.) are applied to create a single vocabulary across event types. Encounter timestamps are merged and DAYS_SINCE_START computed relative to the patient’s first event. A custom `MultimodalTokenizer` assigns integer indices (including <PAD> and <UNK> tokens). The target is the latest care plan code; inputs are all preceding events (preventing data leakage). Time deltas are log-scaled via $\log(1 + t)$.

LSTM Pipeline. Data is organized at the encounter level. Each clinical domain is one-hot encoded and merged into a single wide feature vector per encounter. Static patient demographics (age, marital status, race, ethnicity, gender) are separately encoded. The target is a multi-label binary vector over the top-20 most frequent care plan types from the latest care-plan encounter, with care plan features shifted by one time step to prevent leakage. Sequences are padded to maximum length (capped at 500).

3.3 Model Architectures

Time-Aware Clinical Transformer. An `nn.Embedding` layer maps discrete medical codes to $d_{\text{model}} = 64$ vectors (scaled by $\sqrt{d_{\text{model}}}$). A linear layer projects the scalar log-scaled time delta to d_{model} dimensions; token and time embeddings are summed. A single-layer `TransformerEncoder` ($n_{\text{head}} = 2$) processes the combined embeddings with a key padding mask. The output at the last non-padded position is extracted and passed through a linear classification head over C care plan classes. Training uses cross-entropy loss, Adam ($\text{lr} = 10^{-4}$, weight decay 10^{-4}), gradient clipping (max norm 1.0), dropout $p = 0.3$, and best-model checkpointing over 100 epochs.

Multi-Layer LSTM with Demographics. A linear layer maps the demographic vector to dimension $h = 128$. A 6-layer LSTM ($h = 128$) processes packed encounter sequences; the last valid time step output is extracted. Static and temporal representations are concatenated ($2h = 256$) and

passed through two FC layers (dropout $p = 0.1$) to produce 20 care plan logits. Training uses BCEWithLogitsLoss with class-specific positive weights, Adam (lr = 10^{-3} , weight decay 10^{-3}), ReduceLROnPlateau (patience 5), and early stopping (patience 10) via PyTorch Lightning.

3.4 Evaluation Metrics

The Transformer is evaluated with Top-1/3/5 accuracy, per-class precision/recall/F1, and confusion matrices. The LSTM uses multi-label accuracy (threshold 0.5), macro precision/recall/F1, per-class ROC-AUC curves, and per-class confusion matrices.

3.5 Interpretability

We use Captum’s [9] Integrated Gradients [10] to attribute predictions to individual clinical events. A custom EmbeddingWrapper enables gradient-based attribution on token embeddings. Attributions are computed along the path from a zero-tensor baseline to actual embeddings, then summed across the embedding dimension to yield per-token importance scores. The top- k events by absolute attribution are visualized with positive (green) and negative (red) contributions.

4 Results

4.1 Dataset Statistics

After preprocessing, the Transformer’s unified vocabulary contains $\sim 5,000$ tokens across six clinical domains, while the LSTM’s one-hot encounter vectors yield high-dimensional feature vectors with 20 multi-label targets. Figure 1 shows that observations dominate the record volume (median ≈ 164 per patient) while care plans are sparse (median ≈ 3), and that encounter counts vary widely across patients.

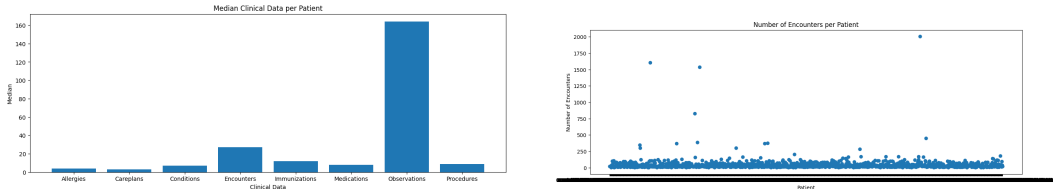


Figure 1: Left: Median clinical records per patient by domain. Right: Distribution of encounter counts per patient.

4.2 Transformer Model Performance

Table 1 summarizes the Transformer’s metrics after 100 epochs with best-model selection on validation loss.

Table 1: Transformer classification metrics on the validation set.

Metric	Score
Top-1 Accuracy	70%
Top-3 Accuracy	91%
Top-5 Accuracy	97%
Macro Precision	0.75
Macro Recall	0.70
Macro F1-Score	0.66

Care plans with more training samples achieve higher precision and recall, while rare plans suffer from limited data. The confusion matrix (Figure 2) shows strong diagonal dominance for common care plan types.

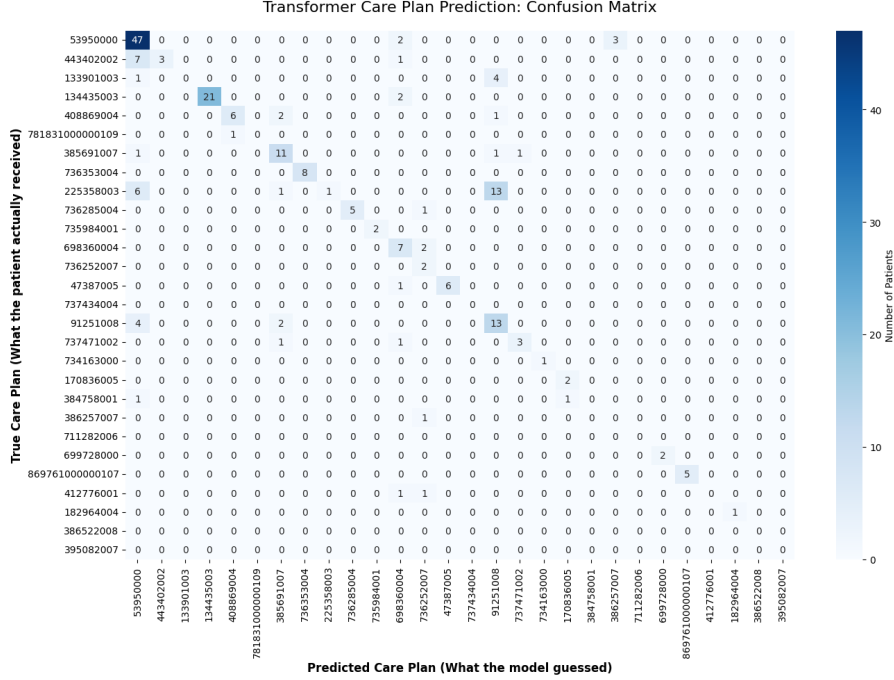


Figure 2: Transformer confusion matrix. Strong diagonal dominance is visible for frequent care plan types (e.g., code 53950000 with 47 correct predictions).

4.3 LSTM Model Performance

Table 2 summarizes the LSTM’s performance with early stopping.

Table 2: LSTM multi-label classification metrics (threshold = 0.5).

Metric	Score
Multi-Label Accuracy	0.88
Macro Precision	0.10
Macro Recall	0.20
Macro F1-Score	0.11

Per-class ROC curves (Figure 3) show the model achieves high AUC for several common care plans (Care Plan 0.99, Cancer 0.98, Dementia management 0.95), while rarer plans exhibit lower discriminative ability. Per-class confusion matrices confirm that for most care plan types the LSTM achieves zero true positives, predicting only the trivial negative class.

4.4 Comparative Analysis

Table 3 presents a direct head-to-head comparison. The Transformer achieves $7.5\times$ higher macro precision, $3.5\times$ higher recall, and $6\times$ higher F1 than the LSTM. The LSTM’s nominally high multi-label accuracy of 0.88 is misleading: with 20 binary targets that are overwhelmingly negative, predicting all-zeros already yields high accuracy.

Key Observations. (1) The Transformer’s self-attention computes direct pairwise scores between all events, preserving long-range dependencies that the LSTM’s sequential hidden state progressively overwrites, especially for patients with >200 encounters. (2) The Transformer’s explicit time embedding via $\log(1 + \Delta t)$ captures irregular visit intervals as a continuous feature, compared to the LSTM’s weaker implicit temporal signal from encounter ordering. (3) The Transformer’s compatibility with Integrated Gradients provides event-level interpretability that the LSTM lacks. (4) Despite having fewer parameters (1 layer, 2 heads vs. 6 LSTM layers), the Transformer achieves

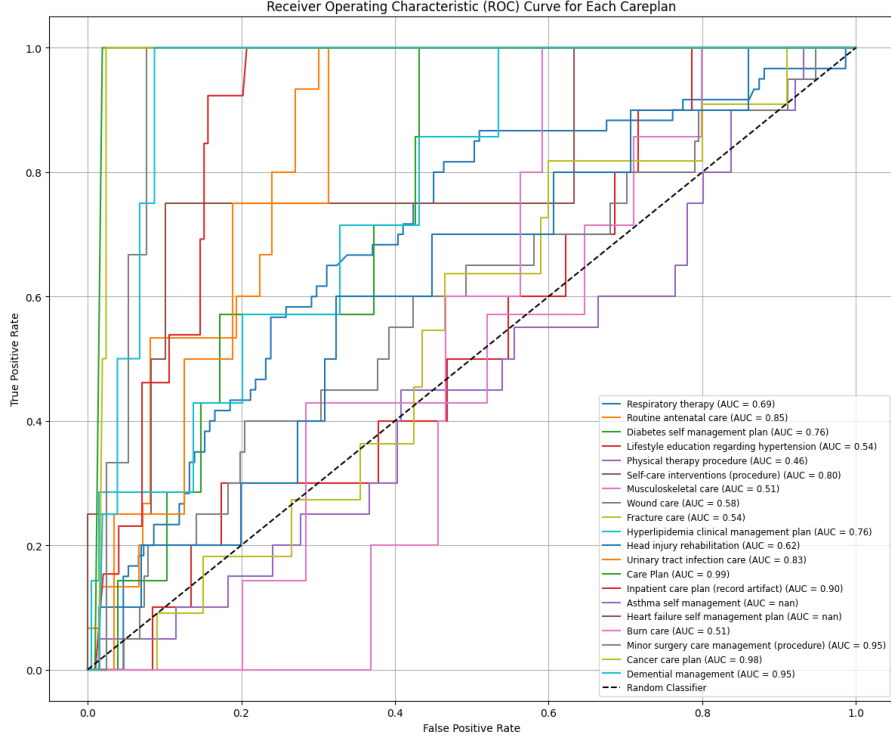


Figure 3: Per-class ROC curves for the LSTM model. Several common care plans achieve high AUC, but rare classes show poor discrimination.

Table 3: Head-to-head metric comparison. The LSTM’s accuracy[†] is inflated by label sparsity.

Metric	Transformer	LSTM
Macro Precision	0.75	0.10
Macro Recall	0.70	0.20
Macro F1-Score	0.66	0.11
Top-1 / Multi-Label Accuracy	70%	88% [†]

[†]Predicting all-zeros yields high accuracy due to sparse labels.

superior performance, confirming that architectural inductive bias matters more than model capacity for this task.

4.5 Interpretability Results

Integrated Gradients attributions on a sample patient reveal that the condition code COND_62106007 (Concussion with no loss of consciousness) at Day 1980 was the strongest positive contributor to the predicted Head Injury Rehabilitation care plan (Figure 4). Secondary drivers include observation scores (QOLS, DALY) at various time points, while routine observations at earlier time points had negative attributions. The model correctly prioritized the acute event over older routine observations, validating that learned attention patterns capture clinically meaningful temporal dynamics.

5 Conclusion and Future Work

We demonstrate that a Time-Aware Clinical Transformer significantly outperforms a Multi-Layer LSTM for care plan prediction from structured FHIR data. The Transformer achieves macro precision/recall/F1 of 0.75/0.70/0.66 compared to the LSTM’s 0.10/0.20/0.11, a gap attributable to self-attention’s ability to preserve long-range clinical dependencies that the LSTM’s sequential hid-

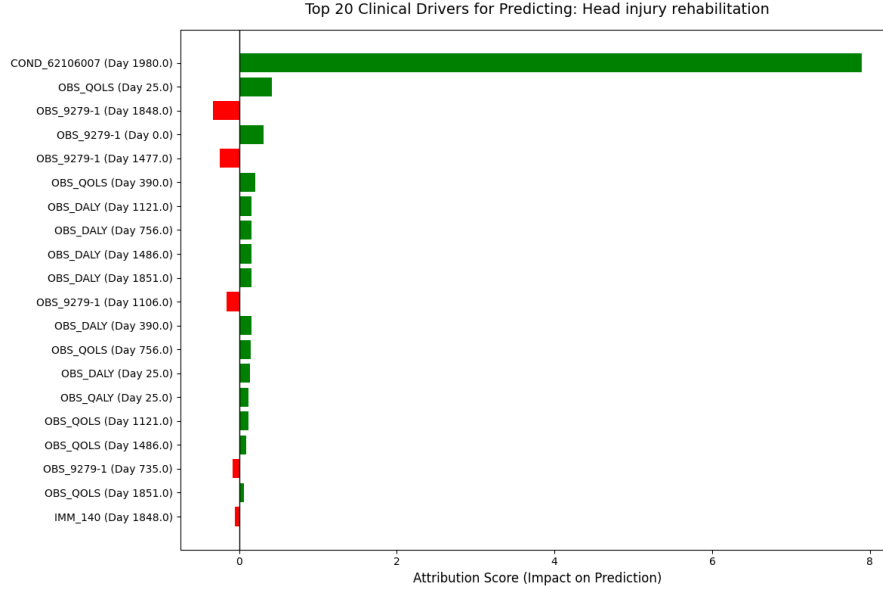


Figure 4: Top 20 clinical drivers for predicting Head Injury Rehabilitation via Integrated Gradients. COND_62106007 (Concussion) at Day 1980 dominates (attribution ≈ 7.9). Green/red bars show positive/negative contributions.

den state progressively overwrites. Integrated Gradients attribution further validates this advantage, showing that the Transformer correctly assigns dominant weight to the clinically relevant acute event (a concussion at Day 1980) when predicting Head Injury Rehabilitation.

Future Work. Key extensions include: (1) ensemble approaches combining the Transformer’s precision with the LSTM’s multi-label capability; (2) pre-training via masked event prediction on larger EHR corpora; (3) validation on de-identified real-world EHR data; (4) multi-modal integration of clinical notes, imaging, and genomic data; and (5) a clinician-facing dashboard presenting care plan recommendations with attribution-based explanations.

Lessons Learned. Careful temporal filtering is essential to prevent data leakage in longitudinal EHR data. Both models require explicit handling of class imbalance, as care plan frequencies follow a long-tailed distribution. Post-hoc interpretability methods serve not only for trust-building but also as model validation, confirming that reasoning aligns with clinical knowledge.

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Project Notebooks

The complete implementation for both models is available as Google Colab notebooks:

- **LSTM Model Notebook:** <https://colab.research.google.com/drive/1wrC9c0FCD5MWrGzBh1btrS4uAr-zValJ?usp=sharing>
- **Transformer Model Notebook:** https://colab.research.google.com/drive/1ytutp_cLkgfuVJONuspkxBZxn02Z1jY0?usp=sharing